

EDUCATIVE COMMENTARY ON ISI 2020 MATHEMATICS PAPERS

(Last Revised on 13/05/2021)

Contents

Introduction	1
Paper 1	3
Paper 2	33
Concluding Remarks	47

Introduction

Commentaries of ISI B.Stat. Entrance Tests began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continue more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, published by Universities Press, Hyderabad.

The additional solution to Q.13 in Paper 1 and the pure geometry proof of the locus in Q.3 of Paper 2 were contributed by Siddhesh Naik, who also pointed out the errors in the original answers to Q.9 of Paper 1 and to Q.7

and Q.8 of Paper 2. Deepanshu Rajvanshi pointed out the mistake in the solution to Q.28 of Paper 1 and many typos. The geometric series solution to Q.6 of Paper 1 was contributed by Ashwni Kumar.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2020 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 For the function on the real line $\mathbb{R}$ given by $f(x) = |x| + |x + 1| + e^x$, which of the following is true?

- (A) It is differentiable everywhere
- (B) It is differentiable everywhere except at $x = 0$ and $x = -1$
- (C) It is differentiable everywhere except at $x = 1/2$
- (D) It is differentiable everywhere except at $x = -1/2$.

Answer and Comments: (B). Call $|x| + |x + 1|$ as $g(x)$ and e^x as $h(x)$. Then $f(x) = g(x) + h(x)$. So f is differentiable at all points where g and h are both differentiable. Further if g is *not* differentiable at some point c , then nor can be f as otherwise we can write g as $f - h$ which would be differentiable at c since h (and hence $-h$) is differentiable everywhere. Hence the problem is reduced to testing where $g(x) = |x| + |x + 1|$ is differentiable. (The crucial point to note is that the sum of a differentiable and non-differentiable function is always non-differentiable. However, the sum of two non-differentiable functions can be differentiable, a trivial example is to take $|x| + (-|x|)$.)

So, after reducing the problem to checking the differentiability of $g(x) = |x| + |x + 1|$, let us write $g(x)$ as $g_1(x) + g_2(x)$ where $g_1(x) = |x|$ and $g_2(x) = |x + 1|$. This time we have to be wary because the sum of two functions both of which are non-differentiable at a common point can be differentiable at that point. But that does not happen here. g_1 is non-differentiable only at 0 and g_2 only at -1 . So their sum, i.e. g is differentiable everywhere except at $x = 0$ and $x = -1$. Hence (B) is true. ■

The non-differentiability of the absolute value function and its variations is quite well-known. It is possible that those who answer the question correctly do so only on the basis of this, without being mindful

of the crucial point in the present problem, viz. that the sum of two functions of which one is differentiable and the other non-differentiable at some point c is also non-differentiable at c . So in that sense the question is not a true test of a candidate's ability to grasp subtlety.

- Q.2 If f, g are real-valued differentiable functions on the real line $\mathbb{R}$ such that $f(g(x)) = x$ and $f'(x) = 1 + (f(x))^2$, then $g'(x)$ equals

(A) $\frac{1}{1+x^2}$ (B) $1+x^2$ (C) $\frac{1}{1+x^4}$ (D) $1+x^4$

Answer and Comments: (A). A simple application of the chain rule to the first equation gives

$$f'(g(x))g'(x) = 1 \tag{1}$$

We are also given that $f'(x) = 1 + (f(x))^2$. (It should have been stated explicitly that this holds for all x .) Replacing x by $g(x)$ we have

$$f'(g(x)) = 1 + (f(g(x)))^2 = 1 + x^2 \tag{2}$$

Substituting this in (1), we get $g'(x) = \frac{1}{1+x^2}$. ■

A simple application of the chain rule. The problem could have been made a little interesting by asking, say, the value of $g(1) - g(0)$. For, one would then need the fundamental theorem of calculus to get the answer as $\pi/4$.

- Q.3 The number of subsets of $\{1, 2, 3, \dots, 10\}$ having an odd number of elements is

(A) 1024 (B) 512 (C) 256 (D) 50

Answer and Comments: (B). There is nothing special about 10 here. We replace it by any positive integer n and call the set $\{1, 2, \dots, n\}$ as S . By induction on n it is easy to show that the number of all subsets of S (including the empty set and the set S itself) is 2^n . For brevity, let us call a subset even or odd according as the number of elements in it is even or odd. We claim that for *every* $n \in \mathbb{N}$, the number of even

subsets of S is 2^{n-1} , i.e. half the number of all subsets of S . This is easy to prove if n is odd, for in that case the complement of every even subset is odd and *vice versa*. So by pairing off every subset with its complement, we see that there are as many even subsets as odd ones. As together their count is 2^n , there are half of each type.

This argument fails if n is even, as in the present problem. But we can apply induction on n . The case $n = 1$ is trivial. For the inductive step, Let $T = \{1, 2, \dots, n-1\}$ and assume that the number of odd subsets of T is 2^{n-2} . This is also the number of even subsets of T . Now classify the odd subsets of S depending on the presence or absence of the element n . Let $A \subset S$ be an odd subset. If $n \in A$, then $A - \{n\}$ is an even subset of T and there are 2^{n-2} such even subsets of S . On the other hand, if $n \notin A$, then A itself is an odd subset of T and there are 2^{n-2} such subsets. Put together the number of odd subsets of S is $2^{n-2} + 2^{n-2} = 2^{n-1}$. By induction, for every $n \in \mathbb{N}$, the number of odd subsets of S is 2^{n-1} .

In the present problem, $n = 10$. So the number of odd subsets of S is $2^9 = 512$. ■

The basic result proved here, viz. that for a set with n elements, the number of odd subsets is 2^{n-1} is fairly well-known and for candidates who know it the answer is instantaneous. Others have to struggle since the short cut which will work for 9 or 11 fails.

Q.4 A group of 64 players in a chess tournament needs to be divided into 32 groups of 2 players each. In how many ways can this be done ?

- (A) $\frac{64!}{32!2^{32}}$ (B) $\binom{64}{2}\binom{62}{2}\cdots\binom{4}{2}\binom{2}{2}$
 (C) $\frac{64!}{32!32!}$ (D) $\frac{64!}{2^{64}}$

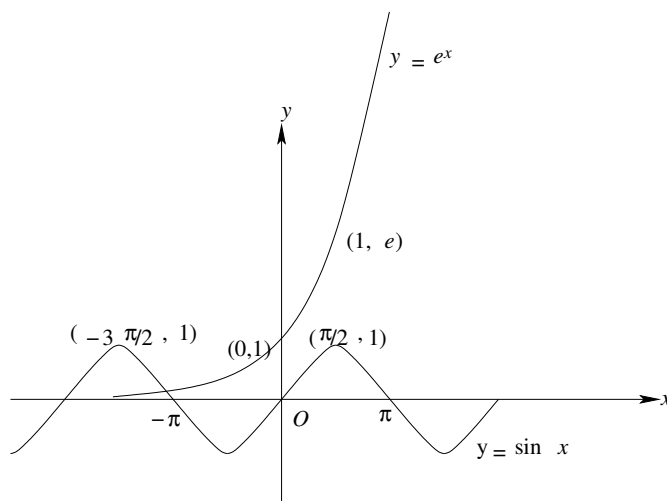
Answer and Comments: (A). Number the pairs from 1 to 32 and assume that they are formed in this order. The first pair can be formed in $\binom{64}{2}$ ways. Now 62 players are left and 2 of them can be chosen to form the second pair in $\binom{62}{2}$ ways. Continuing, the pairs can be

formed in $\binom{64}{2}\binom{62}{2}\cdots\binom{4}{2}\binom{2}{2} = \frac{64!}{2^{32}}$ ways. But the same splitting into pairs appears $32!$ times. So the answer is $\frac{64!}{32!2^{32}}$. ■

Q.5 The number of real solutions of $e^x = \sin(x)$ is

- (A) 0 (B) 1 (C) 2 (D) infinite.

Answer and Comments: (D). If the given equation were $e^x = \cos(x)$, there would be an obvious solution, viz. $x = 0$. As the equation stands, there is no obvious solution. In fact, there cannot be any positive solution because of the inequality $\sin x < x < e^x$ which is valid for all $x > 0$.



But the situation is different for $x \leq 0$. e^x is strictly increasing on $(-\infty, 0]$ and tends to 0 as $x \rightarrow -\infty$. But $\sin x$ assumes all values between -1 and 1 in any interval of length 2π . So the graphs of $y = e^x$ and $y = \sin x$ intersect an infinite number of times for $x \in (-\infty, 0)$. Hence there are infinitely many solutions to the given equation. ■

For a rigorous solution, let $f(x) = \sin x - e^x$. Then for every positive integer n , $f(\frac{(-4n+1)\pi}{2}) > 0$ while $f(\frac{(-4n+3)\pi}{2}) < 0$. Hence by the Intermediate Value Property, f vanishes at least once in each such interval. So there are infinitely many zeros.

Q.6 What is the limit of $\sum_{k=1}^n \frac{e^{-k/n}}{n}$ as n tends to ∞ ?

- (A) The limit does not exist (B) ∞ (C) $1 - e^{-1}$ (D) $e^{-0.5}$

Answer and Comments: (C). We recognise the given sum as a Riemann sum of the function $f(x) = e^{-x}$ for the interval $[0, 1]$ partitioned into n equal parts. Therefore, as $n \rightarrow \infty$, the sum tends to the integral $\int_0^1 f(x)dx = \int_0^1 e^{-x}dx$. Using the Fundamental Theorem of Calculus, the integral equals $-e^{-x}\Big|_0^1 = 1 - e^{-1}$. ■

As the evaluation of the integral is very easy, the crucial step in the solution is to recognise the given sum as a Riemann sum. (Luckily, in this problem, the series can be summed as a geometric series to get the expression as $\frac{1}{n} \frac{1 - e^{-1}}{e^{1/n} - 1}$ whose limit as $n \rightarrow \infty$ is $(1 - e^{-1}) \times \frac{d}{du}(e^u)\Big|_{u=0} = 1 - e^{-1}$.)

At the HSC level, integrals are often not studied as limits of Riemann sums. Those not familiar with them can get the answer by using the additivity and monotonicity properties of integrals. For every positive integer n consider a partition of the interval $[0, 1]$ into n equal parts. Then for $k = 1, 2, \dots, n$, the k -th subinterval of this partition is $[\frac{k-1}{n}, \frac{k}{n}]$. Because of the additivity property of integrals, we have

$$\int_0^1 e^{-x}dx = \sum_{k=1}^n \int_{(k-1)/n}^{k/n} e^{-x}dx \quad (1)$$

The integrand e^{-x} is strictly decreasing on each of these subintervals. So for every $x \in [\frac{k-1}{n}, \frac{k}{n}]$ we have

$$e^{-k/n} \leq e^{-x} \leq e^{-(k-1)/n} \quad (2)$$

Therefore by the monotonicity property of integrals,

$$\int_{(k-1)/n}^{k/n} e^{-k/n}dx \leq \int_{(k-1)/n}^{k/n} e^{-x}dx \leq \int_{(k-1)/n}^{k/n} e^{-(k-1)/n}dx \quad (3)$$

In the first and the third integrals, the integrands are constant and the interval of integration has length $\frac{1}{n}$. Therefore,

$$\frac{e^{-k/n}}{n} \leq \int_{(k-1)/n}^{k/n} e^{-x} dx \leq \frac{e^{-(k-1)/n}}{n} \quad (4)$$

for every $k = 1, 2, \dots, n$. Adding these n inequalities and using (1), we get

$$\sum_{k=1}^n \frac{e^{-k/n}}{n} \leq \int_0^1 e^{-x} dx \leq \sum_{k=1}^n \frac{e^{-(k-1)/n}}{n} \quad (5)$$

The first sum is precisely the sum whose limit as $n \rightarrow \infty$ is to be evaluated. The last sum is also almost this, except that $\frac{e^0}{n}$ is added while $\frac{e^{-1}}{n}$ is missing. Both these expressions tend to 0 as $n \rightarrow \infty$. Therefore as n tends to infinity, the limit, say L , of the first sum is also the limit of the last sum. Hence by the Sandwich theorem the integral in the middle must also equal L . We have already evaluated this integral as $1 - e^{-1}$.

Q.7 Let f, g be differentiable functions on the real line $\mathbb{R}$ with $f(0) > g(0)$. Assume that the set $M = \{t \in \mathbb{R} | f(t) = g(t)\}$ non-empty and $f'(t) \geq g'(t)$ for all $t \in M$. Then which of the following is necessarily true?

- (A) If $t \in M$, then $t < 0$.
- (B) For any $t \in M$, $f'(t) > g'(t)$.
- (C) For any $t \notin M$, $f(t) > g(t)$.
- (D) None of the above.

Answer and Comments: (D). It is easier to deal with a single function rather than a pair of functions. So, as a measure of simplification, we let $h(x) = f(x) - g(x)$. Then the data translates as

- (i) $h(0) > 0$
- (ii) $M = \{t \in \mathbb{R} : h(t) = 0\} \neq \emptyset$ and
- (iii) $h'(t) \geq 0$ for every $t \in M$.

And the first three assertions can be paraphrased as

- (A) If $t \in M$ then $t < 0$, in other words, M contains no positive real numbers.
- (B) $h'(t) > 0$ for every $t \in M$, and
- (C) For every $t \notin M$, $h(t) > 0$

respectively.

Hypothesis (i) rules out the possibility that h is identically zero. For, when this is the case, $M = \mathbb{R}$ and (ii) as well as (iii) hold trivially. But (A), (B) and (C) all fail.

Hypothesis (ii) ensures that h has at least one zero, say t_1 , which is different from 0 if (i) also holds. Then by (iii), $h'(t_1) \geq 0$. If we had $h'(t_1) > 0$, then h would be strictly increasing at t_1 , i.e. for all sufficiently small positive δ ,

$$h(t_1 - \delta) < h(t_1) < h(t_1 + \delta) \quad (1)$$

(This follows from the definition of $h'(t_1)$ as $\lim_{\delta \rightarrow 0} \frac{h(t_1 + \delta) - h(t_1)}{\delta}$ by taking $\epsilon = \frac{h'(t_1)}{2}$ in the definition of a limit.)

If t_1 and $h'(t_1)$ were both positive, then by (1), if $\delta > 0$ is sufficiently small, then $t_1 - \delta > 0$ and $h(t_1 - \delta) < 0$. Since $h(0) > 0$, by the Intermediate Value Property, h would have some zero $t_2 \in (0, t_1 - \delta)$. By (iii), $h'(t_2) \geq 0$. Again if the inequality were strict, repeating the reasoning with t_1 replaced by t_2 , we would get another positive $t_3 \in M$ such that $t_3 < t_2$.

Continuing in this manner, if (iii) were replaced by the stronger hypothesis that $h'(t) > 0$ for all $t \in M$, then h would have no smallest zero. The only way to avoid this is if h has no positive zeros. That means (A) holds.

The catch in this argument is that we are replacing hypothesis (iii) by a stronger one, where $h'(t) > 0$. If all we know is that $h'(t) \geq 0$, it may happen that $h'(t) = 0$. In that case t would be a multiple zero of h . And it may very well happen that $h(t + \delta) < h(t) = 0$ even for some $\delta > 0$. That renders not only (B) but also (C) false, because $t + \delta \notin M$.

Once this point is understood, it is easy to construct an actual counterexample. Cubics are classic examples of functions which are decreasing on both the sides of a critical point, e.g. the cubic $-x^3$. As we want $h(0) > 0$, we modify this and take $h(t)$ as $-(t-1)^3$. Then $h(0) = 1 > 0$. Also h is differentiable everywhere with $h'(t) = -3(t-1)^2$. M equals the singleton set $\{1\}$ and $h'(1) = 0 \geq 0$. So all the conditions (i) to (iii) are satisfied. But none of (A), (B), (C) holds. Hence (D) is true. (In terms of the original functions f and g , we can take $f(t) = h(t) = -(t-1)^3$ and $g(t) = 0$.) ■

There is hardly any computation in the problem. The answer changes dramatically if inequalities are taken as strict. So the question is a good test of fine thinking.

- Q.8 Consider the sequence 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, ... obtained by writing one 1, two 2's three 3's and so on. What is the 2020th term in the sequence ?

(A) 62 (B) 63 (C) 64 (D) 65

Answer and Comments: (C). For every positive integer k , the number of terms which are k or less is $\frac{k(k+1)}{2}$. Call this as T_k . The next $k+1$ terms are equal to $k+1$ each. So the problem reduces to finding the largest positive integer k such that

$$T_k < 2020 \leq T_{k+1} \quad (1)$$

Equivalently k is the largest integer for which

$$k(k+1) < 4040 \quad (2)$$

k can be found systematically by solving the quadratic $k^2 + k - 4040 = 0$ whose roots are $-\frac{1}{2} \pm 2\sqrt{1010}$. $k^2 + k < 4040$ if and only if k lies between these two roots. As k is positive, we discard the negative root and get that $k < -\frac{1}{2} + 2\sqrt{1010}$. Since $31^2 = 961 < 1010 < 1024 = 32^2$, $2\sqrt{1010}$ lies between 62 and 64. Hence $-\frac{1}{2} + 2\sqrt{1010}$ lies between 61.5 and 63.5. Therefore the largest integer k for which the first inequality in (2) holds is either 62 or 63. By easy calculations, $62 \times 63 = 3906 < 4040$ while $63 \times 64 = 4032 < 4040$. Hence $k = 63$. Therefore the 2020-th term of the sequence is 64. ■

The number T_k is called the k -th **triangular number** because it represents the number of dots in a triangle in which the r -th row has r dots for $r = 1, 2, \dots, k$. The problem asks to locate 2020 between two consecutive triangular numbers. Once this basic idea strikes, instead of solving a quadratic equation, one can as well find T_k for $k = 61, 62, 63, 64$ by direct calculations. (Two of them had to be found by calculations anyway.)

- Q.9 Let $A = \{x_1, x_2, \dots, x_{50}\}$ and $B = \{y_1, y_2, \dots, y_{20}\}$ be two sets of real numbers. What is the total number of functions $f : A \rightarrow B$ such that f is onto and $f(x_1) \leq f(x_2) \leq \dots \leq f(x_{50})$?

(A) $\binom{49}{19}$ (B) $\binom{49}{20}$ (C) $\binom{50}{19}$ (D) $\binom{50}{20}$

Answer and Comments: (A). The answer is independent of which particular real numbers appear in the two sets. So we might as well take $y_1, y_2, \dots, y_{20}$ to be the integers from 1 to 20 and further as labels on 20 boxes. Similarly think of $x_1, x_2, \dots, x_{50}$ as balls labeled from 1 to 50. A function f from A to B is a placement of these 50 balls into these 20 boxes. With no restrictions, there are 20^{50} such placements. But here every box is to contain at least one number from 1 to 50. Further the monotonicity requirement translates into saying that each box should contain a block of consecutively labeled balls. So every placement is uniquely determined by the label of the first ball each box contains. The first box must contain the ball labelled 1. For the remaining 19 boxes, the labels must be from 2 to 50 and can be chosen in $\binom{49}{19}$ ways. So this is the total number of functions from A to B which satisfy the two requirements. ■

An excellent combinatorial problem which tests a candidate's ability to grab the essence of a problem and paraphrase it suitably.

- Q.10 The number of complex roots of the polynomial $z^5 - z^4 - 1$ which have modulus 1 is

(A) 0 (B) 1 (C) 2 (D) more than 2.

Answer and Comments: (C). The equation

$$z^5 - z^4 - 1 = 0 \quad (1)$$

can be paraphrased as

$$z^4(z - 1) = 1 \quad (2)$$

Taking absolute values of both the sides and noting that $|z| = 1$ gives

$$|z - 1| = 1 \quad (3)$$

Hence z lies on a circle of radius 1 centred at 1. But z is also given to lie on the unit circle $|z| = 1$. Therefore the two points of intersection of these two circles give the desired complex roots of $z^5 - z^4 - 1$. ■

For the sake of completeness it must be verified that these two points indeed satisfy the polynomial equation. These points come out to be $\alpha = e^{i\pi/3}$ and $\beta = e^{-i\pi/3}$. Note that $\alpha^2 = \omega$ which is a complex cube root of 1. Hence $\alpha^4 = \omega^2$ and $\alpha^5 = -\alpha^2 = -\omega$. Therefore $\alpha^5 - \alpha^4 - 1 = -\omega - \omega^2 - 1$ which equals 0. Hence α is indeed a root of the polynomial $z^5 - z^4 - 1$. Similarly we can check that β is root, although that is hardly necessary because $\beta = \bar{\alpha}$ and the polynomial has real coefficients.

A reasonable problem on complex numbers which ultimately reduces to the sixth roots of unity.

Q.11 The number of real roots of the polynomial

$$p(x) = (x^{2020} + 2020x^2 + 2000)(x^3 - 2020)(x^2 - 2020)$$

is

- (A) 2 (B) 3 (C) 2023 (D) 2025.

Answer and Comments: (B). The paper-setters have been kind by giving $p(x)$ in a factorised form. Otherwise the candidates first have to factor it, using a lot of their precious time and risking that a wrong factorisation may affect the answer.

As it stands, the first factor of $p(x)$ is positive for all real x . The second has only one real root, viz. $\sqrt[3]{2020}$ while the third factor has two real roots $\pm\sqrt{2020}$. Together, $p(x)$ has three real roots. ■

It is hard to think of a more trivial problem. All it needs is that a product vanishes if and only if at least one factor vanishes. To make up, the paper-setters have decorated the problem with the figure 2020, which is the year the examination is held but which has no role in the problem other than it is a positive real number. But this trendy fashion is by now a cliché. It has already been resorted to in Q.8. Two is too many.

- Q.12 Which of the following is the sum of an infinite geometric sequence whose terms come from the set $\{1, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^n}, \dots\}$?

(A) $\frac{1}{5}$ (B) $\frac{1}{7}$ (C) $\frac{1}{9}$ (D) $\frac{1}{11}$

Answer and Comments: (B). Let a be the first term and r the common ratio of the infinite geometric progression chosen from the given sequence. Then its sum, say S , is given by

$$S = \frac{a}{1-r} \quad (1)$$

Here a is of the form $\frac{1}{2^k}$ for some $k \geq 1$ and the common ratio r has to be of the form $\frac{1}{2^s}$ for some $s \geq 1$. Therefore by (1)

$$\begin{aligned} S &= \frac{1/2^k}{1 - (1/2^s)} \\ &= \frac{1}{2^k - 2^{k-s}} \end{aligned} \quad (2)$$

So the problem now is to decide which of the given denominators 5, 7, 9 and 11 can be expressed as $2^k - 2^{k-s}$ for some positive integers k and s . By inspection, we see that 7 can be so expressed by taking $k = 3$ and $s = 3$ too. For a systematic solution, we write $2^k - 2^{k-s}$ as $2^{k-s}(2^s - 1)$. This is a product of a power of 2 and another number which is short by 1 of a power of 2. As all the numbers 5, 7, 9 and 11 are odd, the first factor 2^{k-s} must be 1. That is $k = s$. Hence $2^s - 1$ equals 5, 7, 9 or 11. This is possible only for 7. ■

Problems on sums of geometric series are very common. The present one reduces to a problem in number theory. A simple but innovative problem.

Q.13 The integral part of $\sum_{n=2}^{9999} \frac{1}{\sqrt{n}}$ equals

(A) 196 (B) 197 (C) 198 (D) 199.

Answer and Comments: (B). Call the given sum as S . It has 9998 terms. We add one more term to it to get a sum of 9999 terms which is easier to handle by relating it with a Riemann sum of a suitable function defined over the interval $[1, 10000]$.

$$S + \frac{1}{100} = \sum_{n=2}^{10000} \frac{1}{\sqrt{n}} \quad (1)$$

The interval $[1, 10000]$ is of length 9999. If we divide it into 9999 equal parts then the L.H.S. is a Riemann sum of the function $f(x) = \frac{1}{\sqrt{x}}$ defined over the interval $[1, 10000]$. In the n -th term of this Riemann sum, we are taking the value of $f(x)$ at the rightmost point of the subinterval $[n-1, n]$ for $n = 2, 3, \dots, 10000$. As the integrand is a strictly decreasing function of x , we have

$$S + \frac{1}{100} \geq \int_1^{10000} \frac{1}{\sqrt{x}} dx = 2\sqrt{x} \Big|_1^{10000} = 200 - 2 = 198 \quad (2)$$

Instead of adding $\frac{1}{\sqrt{10000}}$ to S we could as well add $\frac{1}{\sqrt{1}}$. Now we get a Riemann sum which is bigger than the integral because the terms of the sum are the values of $f(x)$ at the leftmost points of the subintervals. That is,

$$S + \frac{1}{10} \leq \int_1^{10000} \frac{1}{\sqrt{x}} dx = 198 \quad (3)$$

Clubbing (2) and (3) together, we get

$$198 - \frac{1}{100} \leq S \leq 198 - \frac{1}{10} \quad (4)$$

Since the numbers on both the ends have the same integral part, viz. 197, we get that S also has 197 as its integral part. ■

This problem has the same basic idea as Q.6 above. Both deal with the integral of a strictly decreasing function. But in Q.6 the problem was to evaluate the limit of a Riemann sum by evaluating the integral. In the present problem, on the other hand, a certain sum is approximated as an integral with an error of not more than 1.

The problem is virtually the same as the second part of Exercise (17.10) except that there the sum whose integral part is to be found is

$$\sum_{n=1}^{10000} \frac{1}{\sqrt{n}}.$$

There is a more elementary, albeit tricky, solution by getting lower and upper bounds on S which are easy to sum by converting them to telescopic series. We begin by noting that

$$\sqrt{n+1} + \sqrt{n} > 2\sqrt{n} > \sqrt{n} + \sqrt{n-1} \quad (5)$$

for every $n \geq 2$. Dividing by 2, taking reciprocals and adding

$$\sum_{n=2}^{9999} \frac{2}{\sqrt{n+1} + \sqrt{n}} < S < \sum_{n=2}^{9999} \frac{2}{\sqrt{n} + \sqrt{n-1}} \quad (6)$$

Rationalising the denominator, the first series becomes $\sum_{n=2}^{9999} 2(\sqrt{n+1} - \sqrt{n}) = 2(\sqrt{10000} - \sqrt{2})$. Similarly, after rationalising the denominators, the last series becomes $\sum_{n=2}^{9999} 2(\sqrt{n} - \sqrt{n-1}) = 2(\sqrt{9999} - \sqrt{1})$. Putting together,

$$200 - 2\sqrt{2} < S < 2\sqrt{9999} - 2 < 200 - 2 = 198 \quad (7)$$

Since $2\sqrt{2} < 3$, the first term is bigger than 197. So from (7) we get $[S] = 197$.

A standard example of a telescopic series is the series $\sum_{k=1}^n \frac{1}{k(k+1)}$, which after rewriting as $\sum_{k=1}^n \frac{1}{k} - \frac{1}{k+1}$ adds instantaneously to $1 - \frac{1}{n+1}$. This artificial example is usually given as an example at the start of the theory of series. Telescopic series rarely occur naturally. But if you can recast a series as a telescopic series no further work is needed to find its sum.

Q.14 Let a_n be the number of subsets of $\{1, 2, \dots, n\}$ that do not contain any two consecutive numbers. Then

- (A) $a_n = a_{n-1} + a_{n-2}$ (B) $a_n = 2a_{n-1}$
 (C) $a_n = a_{n-1} - a_{n-2}$ (D) $a_n = a_{n-1} + 2a_{n-2}$

Answer and Comments: (A). In such problems, it is better to evaluate a_n ‘by hand’ for a few low values of n . Clearly, in the present problem $a_0 = 1$ and $a_1 = 2$ since the requirement of not containing any two consecutive numbers is satisfied vacuously. For a_2 , out of the four subsets of $\{1, 2\}$ we must exclude $\{1, 2\}$ and so $a_2 = 3$. Similarly $a_3 = 5$ since the ‘bad’ subsets are $\{1, 2\}$, $\{2, 3\}$ and $\{1, 2, 3\}$. These simple calculations are enough to rule out (B), (C) and (D). Hence if only one of the given options is correct, it has to be (A).

For an honest solution, classify the ‘good’ subsets of $\{1, 2, \dots, n\}$ into two kinds, depending on the absence or the presence of the element n in them. If S is a good subset of $\{1, 2, \dots, n\}$ and $n \notin S$ then S is a good subset of $\{1, 2, \dots, n-1\}$. There are a_{n-1} good subsets of this type. On the other hand, if $n \in S$, then $n-1 \notin S$ as otherwise S contains two consecutive numbers $n-1$ and n . Hence $S - \{n\}$ is a good subset of $\{1, 2, \dots, n-2\}$. Conversely by adding n to any good subset of $\{1, 2, \dots, n-2\}$, you get a good subset of $\{1, 2, \dots, n\}$ which contains n . As this correspondence is a bijection, there are a_{n-2} good subsets of $\{1, 2, \dots, n\}$ which contain n . Therefore $a_n = a_{n-1} + a_{n-2}$. ■

The relation (A) is the classic **Fibonacci relation**. It occurs in many combinatorial problems. For example, the number of ways to fill an $2 \times n$ rectangle by dominoes (i.e. by 2×1 rectangles) can be shown to be a Fibonacci number. The present problem can be paraphrased as the problem of counting the number of binary sequences (i.e. sequences with only 0 and 1 as possible entries) of length n which contain no two consecutive 1’s and forms the basis of certain probability problems involving tossing a coin repeatedly. (See the JEE 2000 problem in Comment No. 14 of Chapter 22.)

Q.15 There are 128 numbers $1, 2, \dots, 128$ which are arranged in a circular pattern in a clockwise order. We start deleting numbers from this set in a clockwise fashion as follows. First delete the number 2, then

skip the next available number (which is 3) and delete 4. Continue in this manner, that is, after deleting a number, skip the next available number clockwise and delete the number available after that, till only one number remains. What is the last number left ?

- (A) 1 (B) 63 (C) 127 (D) none of the above.

Answer and Comments: (A). We might as well suppose that the numbers are marked consecutively from 0 to 127 and the first round begins with 0 rather than 1. This does not affect the order in which the numbers are deleted, only their labels. In the first round the numbers that get removed are 1, 3, 5, ..., 127 while those remaining are 0, 2, 4, ..., 126. The next round starts at 0 which is not to be removed. In this round all odd multiples of 2 get deleted and the numbers left are 0, 4, 8, ..., 124. The next round begins at 0 and all odd multiples of 4 are removed. Since 128 equals 2^7 , and each round contains half as many numbers as the previous one, in the last round only 0 and 64 are left. Now 0 is not to be removed. 64 is deleted and finally only 0 is left. In the original numbering this is 1. ■

The fact that 128 is a power of 2 simplified the calculations considerably. The general case to get a formula for the last survivor when there are 1 to n numbers is fairly complicated. (See Exercise (24.77).)

- Q.16 Let z and w be complex numbers lying on circles of radii 2 and 3 respectively, with centre $(0, 0)$. If the angle between the corresponding vectors is 60 degrees, then the value of $|z + w|/|z - w|$ is

- (A) $\frac{\sqrt{19}}{\sqrt{7}}$ (B) $\frac{\sqrt{7}}{\sqrt{19}}$ (C) $\frac{\sqrt{12}}{\sqrt{7}}$ (D) $\frac{\sqrt{7}}{\sqrt{12}}$.

Answer and Comments: (A). The problem is more about vectors than complex numbers. This is clear from the second part of the data where the word ‘vector’ is specifically used. But even in the first sentence, z and w could have as well been given as vectors of lengths 2 and 3 respectively. With this change, we are given $|z| = 2, |w| = 3$. Since the angle between them is 60° , we have

$$z \cdot w = |z| \times |w| \times \cos(60^\circ) = 6 \frac{1}{2} = 3 \quad (1)$$

which implies

$$|z + w| = \sqrt{2^2 + 3^2 + 6} = \sqrt{19} \quad (2)$$

By a similar calculation

$$|z - w| = \sqrt{2^2 + 3^2 - 6} = \sqrt{7} \quad (3)$$

Hence the ratio $|z + w|/|z - w|$ equals $\frac{\sqrt{19}}{\sqrt{7}}$. ■

An absolutely simple problem once it is realised that the problem is only about the lengths and the dot product of vectors and that the role of complex numbers is peripheral. Perhaps the intention of the problem was to test the ability to see this. Candidates who miss this and begin by identifying the complex numbers z and w will pay a heavy price in terms of time and possibly accuracy.

- Q.17 Two vertices of a square lie on a circle of radius r and the other two vertices lie on a tangent to this circle. Then the length of the side of the square is

(A) $\frac{3r}{2}$ (B) $\frac{4r}{3}$ (C) $\frac{6r}{5}$ (D) $\frac{8r}{5}$

Answer and Comments: (D). Note that the two vertices that lie on the circle must be adjacent. If they are diagonally apart, then the other two vertices will lie on a chord and not on a tangent. The problem can be tackled with pure geometry. But a coordinate geometry solution is easy too. Take the circle as $\{(x, y) : x^2 + y^2 = r^2\}$ and the two vertices that lie on it as $(\pm \frac{a}{2}, \sqrt{r^2 - \frac{a^2}{4}})$ where a is the side of the square. Then the line, say L containing these two vertices is $y = \sqrt{r^2 - \frac{a^2}{4}}$. The only tangents parallel to L are $y = \pm r$. Since its parallel distance from L is a it cannot be the tangent $y = r$. Therefore it is the tangent $y = -r$ and its distance from L equals the side of the square. This gives the equation

$$a = r + \sqrt{r^2 - \frac{a^2}{4}} \quad (1)$$

Simplifying and squaring gives

$$2ar = \frac{a^2}{4} + a^2 = \frac{5a^2}{4} \quad (2)$$

implying $r = \frac{5a}{8}$ and hence $a = \frac{8r}{5}$. ■

Q.18 For a real number x , let $[x]$ denote the greatest integer less than or equal to x . Then the number of real solutions of $|2x - [x]| = 4$ is

(A) 4 (B) 3 (C) 2 (D) 1.

Answer and Comments: (A). Resolve x as $[x] + \{x\}$ where $\{x\}$ is the fractional part of x , which always lies in the semi-open interval $[0, 1)$. Then $2x - [x] = [x] + 2\{x\}$. Because of the absolute value, the given equation splits into two, viz.

$$[x] + 2\{x\} = 4 \quad (1)$$

$$\text{and } [x] + 2\{x\} = -4 \quad (2)$$

which can be further rewritten as

$$2\{x\} = 4 - [x] \quad (3)$$

$$\text{and } 2\{x\} = -4 - [x] \quad (4)$$

respectively. Note that the right hand sides of both these equations are integers. The L.H.S. $2\{x\}$ takes values in $[0, 2)$. As there are only two integers in this semi-open intervals, (3) gives $0 = 4 - [x]$ and $1 = 4 - [x]$ as two possibilities with $\{x\}$ taking the value 0 and $\frac{1}{2}$ respectively. So (3) has two possible solutions, viz. $x = 4$ and $x = 3.5$.

Similarly (4) leads to either (i) $[x] = -4, \{x\} = 0$ or (ii) $[x] = -3, \{x\} = \frac{1}{2}$. Correspondingly, $x = -4$ or $x = -2.5$. In all, the given equation has 4 solutions. ■

A simple but somewhat repetitious problem. It would have been more interesting if the two equations into which the original equation resolves had some common solution.

Q.19 If $\sin(\tan^{-1}(x)) = \cot(\sin^{-1}(\sqrt{\frac{13}{17}}))$, then x is

$$(A) \frac{4}{17} \quad (B) \frac{2}{3} \quad (C) \sqrt{\frac{17^2-13^2}{17^2+13^2}} \quad (D) \sqrt{\frac{17^2-13^2}{17 \times 13}}.$$

Answer and Comments: (B). Let $\alpha = \tan^{-1}(x)$ and $\beta = \sin^{-1}(\sqrt{\frac{13}{17}})$. Then $\tan \alpha = x$ whence

$$\sin \alpha = \frac{x}{\sqrt{1+x^2}} \quad (1)$$

Similarly, $\sin \beta = \sqrt{\frac{13}{17}}$ which gives

$$\cot^2 \beta = \operatorname{cosec}^2 \beta - 1 = \frac{17}{13} - 1 = \frac{4}{13} \quad (2)$$

As β is in the first quadrant, $\cot \beta$ is positive and so

$$\cot \beta = \frac{2}{\sqrt{13}} \quad (3)$$

The given equation implies

$$\frac{x}{\sqrt{1+x^2}} = \frac{2}{\sqrt{13}} \quad (4)$$

Squaring, and simplifying gives $9x^2 = 4$ whence $x = \pm \frac{2}{3}$. The negative sign is to be discarded since positivity of $\cot \beta$ implies that of $\sin \alpha$ and then of x . ■

A straightforward problem on inverse trigonometric functions. It would have been more interesting if it were so framed that consideration of the sign were mandatory. For example, $-\frac{2}{3}$ could have been given as an option.

Q.20 If the word PERMUTE is permuted in all possible ways and the different resulting words are written down in alphabetical order (also known as dictionary order), irrespective of whether the word has meaning or not, then the 720-th word would be:

(A) EEMPRTU (B) EUTRMPE (C) UTRPMEE (D) MEETPUR.

Answer and Comments: (B). The given word has 7 letters of which one (E) is repeated twice. So the total number of words that can be

formed is $\frac{7!}{2!} = 2520$. The first word would be EEMPRTU, which is option (A). If the first letter is fixed as E, then the number of such words is $6!$ since the remaining 6 letters form a permutation of the distinct symbols P, E, R, M, U and T. There are $6! = 720$ such permutations and the last one among them is UTRPME. So, with the original word PERMUTE, the word EUTRPME comes in the 720-th position. ■

Note that (C) is the last word in the entire list of 2520 words. Although we do not need it, it is instructive to find out where the word in (D), viz. MEETPUR ranks. The approach resembles the way you look up a given word in a dictionary. The only letter in the given word that comes before M is E. So all words beginning with an E come before MEETPUR. The number of such words is $6! = 720$ as we already know. This count includes all words beginning with EE. Next come words beginning with M. Among all such words, those starting with MEE come before any others and the number of such words is $4! = 24$ because there are four letters that remain, viz. T, P, U and R. Out of these, P and R come before T in alphabetical order. So all words beginning with MEEP or MEER come before MEETPUR. There are 6 words of each kind. We have now reached MEET. There are 6 words beginning with MEET. Out of these we are interested only in those that begin with MEETP. There are only two such words, viz. MEETPRU and MEETPUR. Only the former precedes MEETPUR. Hence the words that come before MEETPUR can be classified as (i) those beginning with E, (ii) those beginning with MEEP, (iii) those beginning with MEER, and, (iv) the word MEETPRU. Their counts are 720, 6, 6 and 1. Together 733 words come before MEETPUR. Hence MEETPUR occupies the 734-th position in the alphabetical listing of all the 2520 permutations of the word PERMUTE.

The problem as it stands is simple because the word EUTRPME in (B) happens to be the last one among all those that begin with E and so it is easy to place it without much further work. A more interesting problem would be to ask for the location of a particular word (as we did with MEETPUR) which is not so obvious to locate.

A still more interesting problem would be to give some number, say m , between 1 and 2520 and ask which word comes in the m -th

position. But this is a more challenging problem. The basic idea is similar to that in Q.8, where we had to place a given number between two consecutive triangular numbers. But here it is to be applied repeatedly, so as to trap the given integer m between smaller and smaller blocks of words. To begin with, we identify the end points of the blocks of the permutations starting with a given letter. Thus there are 720 words beginning with E, starting from EEMPRTU and ending with EUTRMPE. They occupy the first 720 places. Next we take words beginning from MEEPRTU and ending with MUTRPEE. This block has length $\frac{6!}{2!} = 360$. Hence the numbers 721 to 1080 lie between these two permutations. Now, if we have $m = 1000$ (say), we partition the second block into several sub-blocks. Thus the first sub-block will run from MEEPRTU to MEUTRPE. It has 120 elements numbered from 721 to 840. The next sub-block is from MPEERTU to MPUTREE with numbers running from 841 to 900. Continuing, we get which permutation corresponds to 1000.

- Q.21 The points $(4, 7, -1), (1, 2, -1), (-1, -2, -1)$ and $(2, 3, -1)$ in $\mathbb{R}^3$ are vertices of a
- (A) rectangle which is not a square.
 - (B) rhombus
 - (C) parallelogram which is not a rectangle
 - (D) trapezium which is not a parallelogram.

Answer and Comments: (C). Call the given points as A, B, C, D respectively. The figures in all four options lie in a plane. But it is stressed in the statement of the problem that the points lie in $\mathbb{R}^3$. So, logically, one should begin by first checking whether these four points indeed lie in a plane. Normally this would require showing that a certain 4×4 determinant vanishes. But in the present problem, that is hardly necessary. By inspection all the points A, B, C, D lie on the plane $z = -1$. In fact, the third coordinate -1 is redundant as far as the problem is concerned. We might as well consider the projections of the four points on the xy -plane. They are $A' = (4, 7), B' = (1, 2), C' = (-1, -2)$ and $D' = (2, 3)$. By trial, we see that the segments $A'C'$ and $B'D'$ have the same midpoint, viz. $(\frac{3}{2}, \frac{5}{2})$. Hence the diagonals $A'C'$

and $B'D'$ bisect each other. Therefore $A'B'C'D'$ is a parallelogram. This rules out option (D). To see which of the remaining three options holds, we first check if $A'C'$ and $B'D'$ are equal in length. A direct computation gives $A'C'$ has length $\sqrt{5^2 + 9^2} = \sqrt{61}$ while the other diagonal $B'D'$ has length $\sqrt{1^2 + 1^2} = \sqrt{2}$. As these are unequal, the parallelogram $A'B'C'D'$ is not a rectangle. The same holds for the quadrilateral $ABCD$. Therefore option (C) is correct. As only one option is given to be correct, we need not check (B), i.e. whether $A'B'C'D'$ is a rhombus. However, we do so for the sake of completeness. The slope of $A'C'$ is $\frac{9}{5}$ while that of $B'D'$ is 1. Hence they are not at right angles, as they ought to be for a rhombus. That rules out option (B). ■

A simple problem once it is realised that the third coordinate of all points is the same and hence plays no role in the problem. The stipulation that the points are in $\mathbb{R}^3$ is more of a mischief than a real hurdle to cross. Those who ignore it are blessed.

Q.22 Let $f(x), g(x)$ be functions on the real line $\mathbb{R}$ such that both $f(x) + g(x)$ and $f(x)g(x)$ are differentiable. Which of the following is FALSE?

- (A) $f(x)^2 + g(x)^2$ is necessarily differentiable.
- (B) $f(x)$ is differentiable if and only if $g(x)$ is differentiable.
- (C) $f(x)$ and $g(x)$ are necessarily continuous.
- (D) If $f(x) > g(x)$ for all $x \in \mathbb{R}$ then $f(x)$ is differentiable.

Answer and Comments: (C). It was mentioned in the solution to Q.1 that the sum of two non-differentiable functions can be differentiable, a trivial counter-example being $f(x) + (-f(x))$ where $f(x)$ is non-differentiable. It was also shown that the sum of a differentiable and a non-differentiable function is always non-differentiable. The contrapositive of this statement is that if $f(x) + g(x)$ is differentiable and one of $f(x)$ and $g(x)$, say $f(x)$, is not differentiable, then the other cannot be differentiable either. In other words, statement (B) is always true. Here we only need that $f(x) + g(x)$ is differentiable. If, in addition, we are given that $f(x)g(x)$ is also differentiable, then $(f(x) + g(x))^2 - 2f(x)g(x)$ is differentiable. Hence statement (A) is also true.

We now come to (C). Since differentiability implies continuity, it is tempting to think that (C) is true. But the catch is that we are *not* given the differentiability of $f(x)$ and $g(x)$ individually, but only of their sum $f(x) + g(x)$ and of their product $f(x)g(x)$. In fact, in the trivial counter-example given in the answer to Q.1, if $g(x) = -f(x)$, then $f(x) + g(x)$ is the constant function 0 which is differentiable everywhere regardless of what $f(x)$ is. If we could so arrange the things that $f(x) + g(x)$ and $f(x)g(x)$ are both constants then we may get an example where $f(x)$ and $g(x)$ are not even continuous. Here a clue is provided by the simple observation that $1 + (-1) = 0$ and $1 \times (-1) = -1$. To see how it works, suppose we can construct two functions $f(x)$ and $g(x)$ each taking only two possible values, viz. 1 and -1 in such a way that where $f(x)$ is 1, the other function $g(x)$ is -1 and vice versa. Then $f(x) + g(x)$ and $f(x)g(x)$ are constant functions 0 and -1 respectively. Hence both are differentiable everywhere, regardless of how $f(x)$ and $g(x)$ behave individually.

Getting such a magic pair of functions which are bad individually but make a perfect couple is easy. Let

$$f(x) = \begin{cases} -1 & \text{if } x \leq 1 \\ 1 & \text{if } x > 1 \end{cases} \quad (1)$$

$$\text{and } g(x) = \begin{cases} 1 & \text{if } x \leq 1 \\ -1 & \text{if } x > 1 \end{cases} \quad (2)$$

Both $f(x)$ and $g(x)$ are continuous everywhere except at 1. So they fit the table. (Many other choices are possible. In fact, if A is any non-empty proper subset of $\mathbb{R}$, we let $B = \mathbb{R} - A$, i.e. the complement of A in $\mathbb{R}$. If we let $f(x)$ equal 1 for $x \in A$ and -1 for $x \in B$, then f has to be discontinuous at any boundary point of A . We can now let $g(x) = -f(x)$.) Hence (C) is false and therefore a right option.

As only one option is given to be correct, it is not necessary to check (D). But, for the sake of completeness, we show that it is a true statement. In (A) we already showed that $f(x)^2 + g(x)^2$ is differentiable. We now subtract $2f(x)g(x)$ which is also differentiable and get that $(f(x)^2 - g(x)^2)$ is differentiable. Call it $u(x)$. Since $f(x) > g(x)$ for all $x \in \mathbb{R}$, we can extract $f(x) - g(x)$ as the unique positive square root $\sqrt{u(x)}$ of $u(x)$ which is always positive. The square root function is

differentiable on the positive u -axis. So, by the chain rule, $\sqrt{u(x)}$, i.e. $f(x) - g(x)$ is differentiable. ■

An unusual problem, a good test of fine thinking and the ability to construct counter-examples.

- Q.23 Let S be the set consisting of all those real numbers that can be written as $p - 2a$ where p and a are the perimeter and the area of a right-angled triangle having base length 1. Then S is

(A) $(2, \infty)$ (B) $(1, \infty)$ (C) $(0, \infty)$ (D) the real line $\mathbb{R}$.

Answer and Comments: (A). Let x be the height of the triangle. Then its perimeter p is $1 + x + \sqrt{1 + x^2}$, while its area a is $\frac{1}{2}x$. Hence $p - 2a = 1 + \sqrt{1 + x^2}$. Here x varies from 0 to ∞ and as it does so, $\sqrt{1 + x^2}$ takes all values in the interval $(1, \infty)$. So, $p - 2a = 1 + \sqrt{1 + x^2}$ takes all values in $(2, \infty)$. ■

An absolutely simple problem, little beyond the Pythagoras theorem.

- Q.24 Let $S = \{1, 2, \dots, n\}$. For any non-empty subset A of S , let $l(A)$ denote the largest number in A . If $f(n) = \sum_{A \subseteq S} l(A)$, that is, $f(n)$ is the sum of the numbers $l(A)$ while A ranges over all the non-empty subsets of S , then $f(n)$ is

(A) $2^n(n + 1)$ (B) $2^n(n + 1) - 1$
(C) $2^n(n - 1)$ (D) $2^n(n - 1) + 1$.

Answer and Comments: (D). As in Q.14, it is better to find the answer ‘by hand’ for some lower values of the integer n . Let us take $n = 3$. Then $\{1, 2, 3\}$ has 7 non-empty subsets, viz. three singletons, three doubletons and the entire set S . The contribution to $f(n)$ from these three classes are $1 + 2 + 3$, $2 + 3 + 3$ and 3. So, $f(3) = 6 + 8 + 3 = 17$. This is an odd number and that rules (A) and (C) out. By a direct calculation, the expression in (B) is 33 for $n = 3$. So (B) is also incorrect. That leaves only (D).

For an honest solution, we classify the non-empty subsets of S according to their largest elements. For a subset A of $S = \{1, 2, \dots, n\}$,

the largest element, say r , can be any integer from 1 to n . For a given r , r will be the largest element of a subset A if and only if $r \in A$ and $A - \{r\}$ is a (possibly empty) subset of $\{1, 2, \dots, r\}$. As there are 2^{r-1} such subsets, we see that the contribution to $f(n)$ from subsets A for which $l(A) = r$ is $2^{r-1}r$. Hence

$$f(n) = \sum_{r=1}^n 2^{r-1}r \quad (1)$$

There are various ways to show that this sum equals the option in (D), viz. $2^n(n-1) + 1$. The easiest is by induction on n . The case $n = 1$ is trivially true. For the inductive step, assume that

$$f(k) = 2^k(k-1) + 1 \quad (2)$$

Note that $f(k+1)$ is obtained from $f(k)$ by merely adding one more term, viz. $2^k(k+1)$. Therefore

$$\begin{aligned} f(k+1) &= f(k) + 2^k(k+1) \\ &= 2^k(k-1) + 1 + 2^k(k+1) \\ &= 2^k(2k) + 1 \\ &= 2^{k+1}k + 1 \\ &= 2^{k+1}(k+1-1) + 1 \end{aligned} \quad (3)$$

Hence the truth for k implies that for $k+1$. Therefore (D) is true for every positive integer n . ■

A fairly good problem. But the wrong options can be eliminated too easily with just one value of n , viz. 3. (Even $n = 2$ will do.) So the question is ill suited for a multiple choice test.

Q.25 If a, b, c are distinct odd natural numbers, then the number of rational roots of the polynomial $ax^2 + bx + c$

- (A) must be 0.
- (B) must be 1.
- (C) must be 2.
- (D) cannot be determined from the given data..

Answer and Comments: (A). Suppose $x = \frac{p}{q}$ is a rational root of $ax^2 + bx + c$ where p, q are integers. Without loss of generality, we assume that p and q are relatively prime, i.e. have no common factor other than ± 1 . Then a direct substitution and a little simplification gives

$$ap^2 + bpq + cq^2 = 0 \quad (1)$$

As p divides the first two terms, it must divide the third, i.e. cq^2 . But p is relatively prime to q . Hence p must divide c . But then p is odd because c is odd. By a similar reasoning we get that q is odd. But then all the three terms on the L.H.S. of (1) are odd. Hence their sum is also odd and cannot be 0. So the polynomial $ax^2 + bx + c$ has no rational roots. ■

A simple problem which is more about number theory than quadratic polynomials. The hypothesis that a, b, c are all odd is needed crucially. But their distinctness is nowhere needed.

The reasoning adopted also applies to prove a more general result that if $\frac{p}{q}$ is a rational root (in its reduced form) of a polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + a_1 x + a_0 \quad (2)$$

where $a_0, a_1, \dots, a_n$ are all integers then p divides a_0 and q divides a_n . This simple result is called the **rational root test**. Since a_n and a_0 have only a finite number of divisors each, we can identify all possible rational roots of $p(x)$. Trying them one by one, we get all rational roots of $p(x)$. As a numerical example, suppose $p(x) = 8x^3 - 4x^2 - 4x + 1$. Then the only possible rational roots of $p(x)$ are $\pm 1, \pm 2, \pm 4$ and ± 8 . (If we make a clever substitution $y = 2x$, then the polynomial becomes $y^3 - 2y^2 - 2y + 1$ and its only possible rational roots are ± 1 both of which are ruled out by direct substitution.)

Despite its simplicity, the rational root test is a useful result, because when $n > 2$, there are no easy formulas for the roots of $p(x)$. Sometimes we can guess some root, say α , and consequently a factor $x - \alpha$ of $p(x)$. This passes the problem to the other factor of $p(x)$ which is a polynomial of one degree smaller. But even guessing one root is often not easy. The rational root test shortlists the search as far as

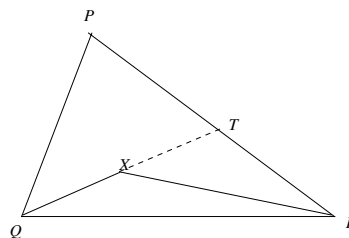
rational roots are concerned. It may, of course, happen that there are no rational roots even when all the coefficients of $p(x)$ are integers. In that case the test is of little help and all one can do is to get the roots approximately.

Q.26 Let A, B, C be finite subsets of the plane such that $A \cap B, B \cap C$ and $C \cap A$ are all empty. Let $S = A \cup B \cup C$. Assume that no three points of S are collinear and also assume that each of A, B, C has at least 3 points. Which of the following statements is always true ?

- (A) There exists a triangle having a vertex from each of A, B, C that does not contain any point of S in its interior.
- (B) Any triangle having a vertex from each of A, B, C must contain a point of S in its interior.
- (C) There exists a triangle having a vertex from each of A, B, C that contains all the remaining points of S in its interior.
- (D) There exist 2 triangles, both having a vertex from each of A, B, C such that the two triangles do not intersect.

Answer and Comments: (A). We recall a fact from school geometry.

Suppose X is an interior point of a triangle PQR . Then $PQ + PR > XQ + XR$. For a proof, produce QX to meet PR at T . Then $PQ + PT > QT = QX + XT$ and $XT + TR > XR$. The result follows by adding these two inequalities and canceling XT from both the sides.



To put this fact to use, consider the set of all triangles PQR with $P \in A, Q \in B$ and $R \in C$. There are only finitely many such triangles. So we may choose PQR to have the least perimeter among all such triangles. We claim that there is no point of S which lies in the interior of PQR . Let, if possible, X be such a point. Since $S = A \cup B \cup C$, X lies in one of the three sets A, B, C . Without loss of generality, assume that $X \in A$. Then by the fact mentioned above, the triangle XQR has a smaller perimeter than the triangle PQR . This contradicts the

minimality of the perimeter of PQR . This contradiction proves that PQR contains no interior point of S . ■

Although the proof is simple, once it is realised how to apply the fact from geometry given above, there is an important principle involved here, called the **extremality principle**. In this method, the object whose existence is to be proved is identified as a point in some set (in our case, the set of all triangles with one vertex in each of A, B, C) at which some real valued function (in our case, the perimeter) attains an extremum (in our case the minimum). One of the solutions to Q.5 in the second paper ISI 2019 was also based on the extremality principle.

Q.27 Shubhaangi thinks she may be allergic to Bengal gram and takes a test that is known to give the following results:

- For people who really do have the allergy, the test says 'Yes' 90% of the time.
- For people who do not have the allergy, the test says "Yes" 15% of the time.

If 2% of the population has the allergy and Shubhaangi's test says "Yes", then the chances that Shubhaangi does really have the allergy are

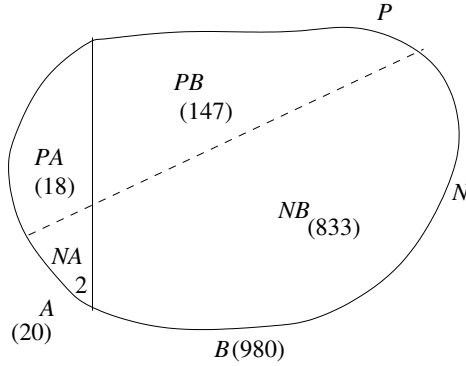
- (A) $1/9$
- (B) $6/55$
- (C) $1/11$
- (D) cannot be determined from the given data.

Answer and Comments: (B). This is a problem on conditional probability, viz. to find the probability that Shubhaangi has the allergy, given that she tests positive. (We assume that the test has only two possible outcomes, positive and negative, so that their probabilities are complementary, regardless of whether they are conducted on persons having the allergy or not having the allergy, even though the values of each are different for the two classes.)

The data is in terms of percentages. We can directly convert them to probabilities. But we can also convert the percentages to

actual numbers by assuming some (arbitrary) numerical size for the sample space. This has the advantage that the data can be described more visually in terms of a suitable Venn diagram. To avoid dealing with fractional numbers of persons, we take the population size to be 1000. We divide it into two classes, those who are allergic and their complement. We denote them by A and B respectively, separated by a vertical line. (The Venn diagrams are only indicative and the areas of various regions are generally not proportional to their numerical sizes.) As the population size is taken as 1000, the last piece of data implies

$$|A| = 20 \quad \text{and} \quad |B| = 1000 - 20 = 980 \quad (1)$$



Next we divide the population into two mutually disjoint subsets P and N representing those which test positive and negative respectively. They are divided by a dotted line. We show the intersections of the form $X \cap Y$ simply by XY . Thus PA is the subset of those who are allergic and also test positive. It has $20 \times \frac{90}{100} = 18$ elements. Hence its complement in A , viz, $N \cap A = NA$ has $|A| - |PA| = 20 - 18 = 2$ elements. Similarly we calculate the cardinality of PB as $980 \times \frac{15}{100} = 147$ and finally, $|NB| = |N| - |NA| = 980 - 147 = 833$ as shown in the Venn diagram above.

The desired probability is the ratio $\frac{|PA|}{|P|}$. We already know $|PA| = 18$. Also

$$|P| = |PA| + |PB| = 147 + 18 = 165 \quad (2)$$

Hence the probability that Shubhaangi has the allergy given that she tests positive for it is $\frac{18}{165} = 6/55$. ■

The calculations would have been essentially the same had we worked in terms of conditional probabilities. But the Venn diagrams make the presentation graphic and easier to understand. The major limitation of Venn diagrams is that they are planar. So they can handle only two ways of dividing the sample space (or, what is formally called a 2×2 contingency). If there were a third parameter, e.g. the sex of the person tested which affected some of the probabilities, the problem would be more complex and cannot be handled by planar Venn diagrams.

- Q.28 For any real number x , let $[x]$ be the greatest integer m such that $m \leq x$. Then the number of points of discontinuity of the function $g(x) = [x^2 - 2]$ on the interval $(-3, 3)$ is

(A) 5 (B) 9 (C) 13 (D) 16.

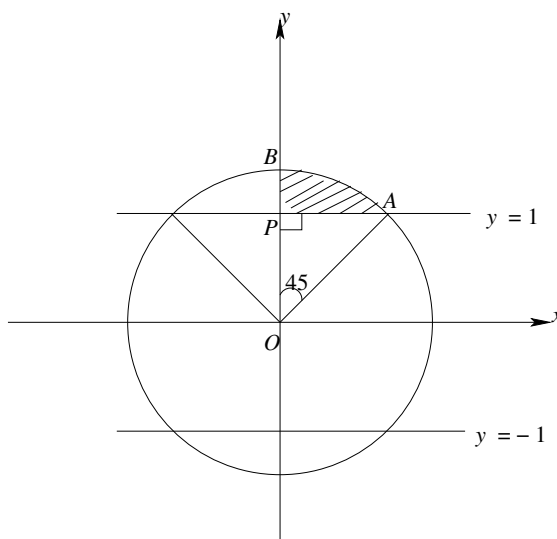
Answer and Comments: (D). The function $x^2 - 2$ is continuous everywhere. So to test the continuity of the the composite function $[x^2 - 2]$, we only need to check where the integral part function is discontinuous. This happens at all integer points. For $x \in (-3, 3)$, $x^2 - 2 \in [-2, 7)$. So the possible points of discontinuity of $g(x) = [x^2 - 2]$ are those where $x^2 - 2 = -2, -1, 0, 1, 2, 3, 4, 5$ and 6 . Correspondingly, $x^2 = 0, 1, 2, 3, 4, 5, 6, 7, 8$. Hence the possible points of discontinuity are $x = 0, \pm 1, \pm\sqrt{2}, \pm\sqrt{3}, \pm 2, \pm\sqrt{5}, \pm\sqrt{6}, \pm\sqrt{7}$ and $\pm\sqrt{8}$. But we must discard 0 since x^2 and hence $x^2 - 2$ are even functions of x . So, on both the sides of 0 , if x is near 0 , then $[x^2 - 2] = -2$. Hence, in all there are 16 points at which $g(x)$ is discontinuous. ■

A simple problem whose basic idea is that the integral part function is discontinuous at all integer points. Care is to be taken to exclude the end points ± 3 .

- Q.29 The area of the region in the plane $\mathbb{R}^2$ given by the points (x, y) satisfying $|y| \leq 1$ and $x^2 + y^2 \leq 2$ is

(A) $\pi + 1$ (B) $2\pi - 2$ (C) $\pi + 2$ (D) $2\pi - 1$.

Answer and Comments: (C). The region, say R , lies inside the disc of radius $\sqrt{2}$ centred at the origin $(0,0)$ and between the lines $y = 1$ and $y = -1$.



The area of the disc is 2π . From this we have to subtract four times the shaded area to get the area of R . The shaded area is the difference between the area of the sector OAB and the area of the right angled triangle OPA with hypotenuse $\sqrt{2}$. The former is $\frac{2\pi}{8} = \frac{\pi}{4}$, while the latter is $\frac{1}{2}$. Hence the shaded area is $\frac{\pi-2}{4}$. Subtracting 4 times this from 2π we get the area of R as $2\pi - \pi + 2 = \pi + 2$. ■

A very straightforward problem. The calculations are simple too.

Q.30 Let n be a positive integer and $t \in (0,1)$. Then $\sum_{r=0}^n r \binom{n}{r} t^r (1-t)^{n-r}$ equals

- (A) nt (B) $(n-1)(1-t)$ (C) $nt + (n-1)(1-t)$ (D) $(n^2 - 2n + 2)t$.

Answer and Comments: (A). The first term of the summation (where $r = 0$) is redundant. But it is kept because if we drop the coefficient r then the sum becomes $\sum_{r=0}^n \binom{n}{r} t^r (1-t)^{n-r}$ where the first term is *not*

redundant. In fact, by the binomial theorem, this sum equals $(t+1-t)^n$ which is simply the constant 1. The presence of the nagging coefficient makes the problem more difficult and hence interesting.

If the coefficient were some constant (independent of r) we could take it out. There is a way to arrange this using the following combinatorial identity

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1} \quad (1)$$

which can be proved simply by writing the L.H.S. as $\frac{n!}{r!(n-r)!}$ and further as $\frac{n}{r} \frac{(n-1)!}{(r-1)!((n-1)-(r-1))!}$. Hence $r \binom{n}{r} = n \binom{n-1}{r-1}$ for $r \geq 1$. Dropping the term for which $r = 0$, we can now take nt common and with a change of index (replacing $r-1$ by k), the sum becomes $nt \sum_{k=0}^{n-1} \binom{n-1}{k} t^k (1-t)^{n-k}$ which equals nt by the binomial theorem. ■

The use of the binomial theorem is marginal. The crucial step in the solution is the identity (1).

ISI BStat-BMath-UGA-2020 Paper 2 with Comments

Q.1 Let i be a root of the equation $x^2 + 1 = 0$ and let ω be a root of the equation $x^2 + x + 1 = 0$. Construct a polynomial

$$f(x) = a_0 + a_1x + \dots + a_nx^n$$

where $a_0, a_1, \dots, a_n$ are all integers such that $f(i + \omega) = 0$.

Solution and Comments: Let

$$\alpha = i + \omega \tag{1}$$

We have to find a polynomial with integer coefficients which has α as a root. The numbers i and ω both have this property, although the corresponding polynomials are different. Specifically, we have

$$i^2 = -1 \tag{2}$$

$$\text{and } \omega^2 + \omega = -1 \tag{3}$$

So, we not only want to eliminate i and ω in (1), (2) and (3), but we also want that the resulting equation in α is a polynomial equation with integer coefficients.

We have a choice. We can first eliminate one of i and ω so as to express the other one in terms of α and then use the equation satisfied by it. We choose to eliminate ω first. Since $\omega = \alpha - i$, from (3) we get

$$(\alpha - i)^2 + (\alpha - i) + 1 = 0 \tag{4}$$

Expanding and simplifying, we get

$$\alpha^2 + \alpha = (2\alpha + 1)i \tag{5}$$

To eliminate i we simply square both the sides and use (2). Then

$$(\alpha^2 + \alpha)^2 = -(2\alpha + 1)^2 \tag{6}$$

Collecting all the terms in α on one side we get

$$\alpha^4 + 2\alpha^3 + 5\alpha^2 + 4\alpha + 1 = 0 \tag{7}$$

So, we can take $f(x)$ as $1 + 4x + 5x^2 + 2x^3 + x^4$. ■

We have shown something more than is asked. Not only have we constructed a polynomial with integer coefficients which has $i + \omega$ as a root, we have constructed such a polynomial which is **monic**, i.e. whose leading coefficient is 1. A number is called an **algebraic integer** if it is a root of a monic polynomial with integer coefficients. Thus all integers as well as both i and ω are algebraic integers. But $\frac{1}{2}$ is not. The present problem is a special case of a general result that the sum of two algebraic integers is an algebraic integer. Of course, the general result only shows the existence of a desired monic polynomial. Here we have actually constructed it for $i + \omega$,

Q.2 Let a be a fixed real number. Consider the equation

$$(x + 2)^2(x + 7)^2 + a = 0, x \in \mathbb{R},$$

where $\mathbb{R}$ is the set of real numbers. For what values of a , will the equation have exactly one double root ?

Solution and Comments: The given equation is a polynomial equation of the form

$$f(x) = 0 \tag{1}$$

where

$$f(x) = (x + 2)^2(x + 7)^2 + a \tag{2}$$

If a real number α is a double root of $f(x)$, then not only $f(\alpha) = 0$, but $f'(\alpha) = 0$, because $f(x)$ can be written as $(x - \alpha)^2 g(x)$ for some polynomial $g(x)$ of degree 2 lower than $f(x)$. So, if α is a double root of (1), then we have

$$(\alpha + 2)^2(\alpha + 7)^2 + a = 0 \tag{3}$$

$$\text{and } 2(\alpha + 2)(\alpha + 7)(2\alpha + 9) = 0 \tag{4}$$

The last equation implies that the only possible values of α are $-2, -7$ and $-\frac{9}{2}$. For $\alpha = -2$, (3) implies that $a = 0$. But in that case both -2

and -7 would be double roots of $f(x)$. As we want only those values of a for which $f(x)$ has only one double root, $\alpha = -2$ is not permissible. By a similar reasoning, $\alpha = -7$ is not permissible. So we try the third possible value of α , viz. $\alpha = -\frac{9}{2}$. Since (3) is to hold for this value of α , we get $a = -(\frac{5}{2})^4 = \frac{625}{16}$. So this is the only value of a for which $f(x)$ has a unique double root. ■

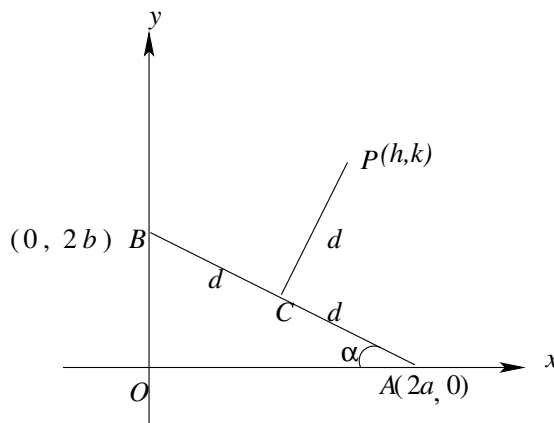
A good problem based on characterisation of a double root of a polynomial as a common root of the polynomial and of its derivative.

Q.3 Let A and B be variable points on the x -axis and y -axis respectively such that the line segment AB is in the first quadrant and of a fixed length $2d$. Let C be the midpoint of AB and P be a point such that

- (a) P and the origin are on the opposite sides of AB and
- (b) PC is a line segment of length d which is perpendicular to AB .

Find the locus of P .

Solution and Comments: The data is shown in the figure below. Evidently the length of AB is given to be $2d$ rather than d to avoid messy fractions. So we too let $A = (2a, 0)$ and $B = (0, 2b)$. Then $C = (a, b)$ and $d = \sqrt{a^2 + b^2}$. Also a, b are both positive as the segment AB is given to lie in the first quadrant.



Let (h, k) be the current coordinates of the point P . We are given that $CP = d$. If we can find the slope of CP , we can determine h and k .

Note that the slope of AB is $-\frac{b}{a}$. Hence the slope of CP is $\frac{a}{b}$. As P lies on the opposite side of AB , we get

$$h = a + \frac{bd}{\sqrt{a^2 + b^2}} = a + b \quad (1)$$

$$\text{and } k = b + \frac{ad}{\sqrt{a^2 + b^2}} = a + b \quad (2)$$

So $h = k$ and therefore the point P lies on the line $y = x$. It is tempting to think, therefore, that this line is the locus of P . But not all points on this line can be in the locus. For example, O is not in it, as it is specifically given that P and O lie on the opposite side of AB .

To determine exactly which points on the line $y = x$ are in the locus, we let $\alpha = \angle OAB$ shown in the figure above. Then $a = d \cos \alpha$ and $b = d \sin \alpha$. Therefore $h = d(\sin \alpha + \cos \alpha)$. d is fixed. So the problem reduces to finding the range of the function $\sin \alpha + \cos \alpha$ as α varies from 0 to $\frac{\pi}{2}$. This can be done either using calculus or by writing $\sin \alpha + \cos \alpha$ as $\sqrt{2}(\cos \frac{\pi}{4} \cos \alpha + \sin \frac{\pi}{4} \sin \alpha)$ i.e. as $\sqrt{2} \cos(\frac{\pi}{4} - \alpha)$ which is maximum when $\alpha = \frac{\pi}{4}$ and minimum when $\alpha = 0$ or $\frac{\pi}{2}$. Hence the point $P(h, h)$ lies on the segment from (d, d) to $(\sqrt{2}d, \sqrt{2}d)$. ■

The problem is an odd combination of two parts. The first part is purely geometric and the answer is unexpectedly simple. (For a pure geometry proof, note that the quadrilateral $OBPA$ is cyclic as two opposite angles are 90° . Hence $\angle BOP = \angle BAP = 45^\circ$.) The second part consists of finding the maximum and the minimum of a trigonometric function. In the past, finding the locus of a point P was taken merely to identify some curve on which P always lies. Nowadays, it is expected to identify the subset, say S , of the curve which consists precisely of those points which P can occupy and no others.

Q.4 Let a real-valued sequence $\{x_n\}_{n \geq 1}$ be such that

$$\lim_{n \rightarrow \infty} nx_n = 0.$$

Find all possible real values of t such that $\lim_{n \rightarrow \infty} x_n (\log n)^t = 0$.

Solution and Comments: We assume that x_n is non-negative as otherwise we replace x_n by $|x_n|$ which affects neither the hypothesis

nor the conclusion. The given condition means that as $n \rightarrow \infty$, x_n tends to 0 faster than $\frac{1}{n}$.

So, for all t for which $(\log n)^t$ tends to ∞ slower than n as $n \rightarrow \infty$, we would have $\lim_{n \rightarrow \infty} \frac{(\log n)^t}{n} = 0$ which would imply that $\lim_{n \rightarrow \infty} x_n (\log n)^t = 0$ because $|x_n (\log n)^t| < |x_n| n$ for all sufficiently large n .

Now $\log n$ tends to ∞ as n tends to ∞ . But it does so slower than n and, in fact, than any positive power of n . That is, for every positive α , no matter how small, $\lim_{n \rightarrow \infty} \frac{\log n}{n^\alpha} = 0$. (A proof of this can be given using the $\frac{\infty}{\infty}$ form of the l'Hôpital's rule.)

We now address the given problem. First, assume $t > 0$. Write $\frac{(\log n)^t}{n}$ as $\left(\frac{\log n}{n^{1/t}}\right)^t$. The parenthesized expression tends to 0 by taking $\alpha = \frac{1}{t}$ in what we said above. But then so does its t -th power since $t > 0$. If $t = 0$, then $x_n (\log n)^t = x_n$ which tends to 0 as n tends to ∞ since $|x_n| < \frac{1}{n}$ for all sufficiently large n . If $t < 0$, then also there is no problem because $(\log n)^t = \frac{1}{(\log n)^{-t}} \rightarrow 0$ as $n \rightarrow \infty$. So certainly the product $x_n (\log n)^t$ tends to 0 as $n \rightarrow \infty$. (Here all we need is that $\{x_n\}$ is bounded.)

Thus, for all real values of t , $x_n (\log n)^t \rightarrow 0$ as $n \rightarrow \infty$. ■

The essential idea in the solution is merely that as $n \rightarrow \infty$, $\log n$ tends to ∞ slower than any positive power n^α of n . Verbally, this is often expressed by saying that logarithmic growth is smaller than algebraic growth. An analogous comparison is that the algebraic growth is slower than the exponential growth. In precise terms, any power and hence any polynomial in n tends to ∞ slower than any exponential a^n where the base a is greater than 1. Thus, in simple interest, the total grows linearly while in compound interest it grows exponentially. Such comparisons are also relevant in dealing with the spread of some pandemic such as covid 19.

Q.5 Prove that the largest pentagon (in terms of area) that can be inscribed in a circle of radius 1 is regular (i.e., has equal sides).

Solution and Comments: Let O be the centre of the circle and V_1, V_2, V_3, V_4 and V_5 the vertices (in counter-clockwise order). Let θ_i be the angle between OV_i and OV_{i+1} for $i = 1, 2, 3, 4, 5$ with the understanding that V_6 is to be treated as V_1 . Then the area, say A of the pentagon is given by

$$A = \frac{1}{2}(\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4 + \sin \theta_5) \quad (1)$$

We also have

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 = 2\pi \quad (2)$$

The problem amounts to showing that the maximum of A subject to the constraint (2), occurs when all θ 's are equal. There is an advanced method for this, called the method of **Lagrange's multipliers**. It requires the concept of partial derivatives. Briefly, the function A in (1) which is to be maximised (or minimised) is called the **objective function** and the function on the L.H.S. of (2), which restricts these variables (so that they are not independent) is called a **constraint function**. In essence the method says that at a point $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ where the extremum occurs, the partial derivatives of the objective function are proportional to those of the constraint function. Thus differentiating (1) and (2) partially w.r.t. each of the θ_i 's, we get that there is some constant (called the multiplier) λ such that

$$\frac{1}{2} \cos \theta_i = \lambda 1 \quad (3)$$

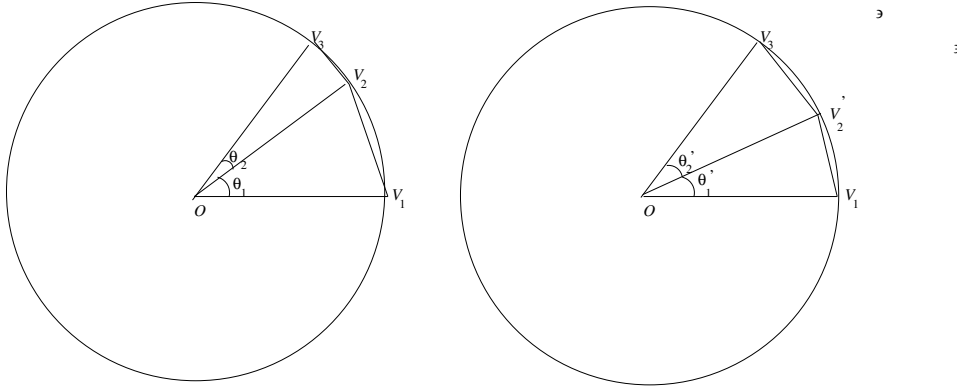
for $i = 1, 2, 3, 4, 5$. Since all θ 's lie in $[0, 2\pi]$ and they add to 2π , it is not hard to show that the only way this can happen is when all θ 's are equal, i.e. when the pentagon is regular.

This method can also be applied to many problems of triangular optimisation (see Chapter 14). In fact, it makes a mincemeat of such problems. But the method is well beyond the HSC level. So, we give another solution which is elementary and based on what may be called the principle of marginal improvement. Specifically, we shall show that if $\theta_1 \neq \theta_2$, then we can get another pentagon with a higher area. The same reasoning will apply to show that for a pentagon of maximum area, all adjacent angles are equal and hence the pentagon is regular.

To show that $\theta_1 = \theta_2$, assume that they are unequal. We shall replace the vertex V_2 by another vertex V'_2 , leaving all other vertices intact so that the pentagon with vertices V_1, V'_2, V_3, V_4 and V_5 has higher area than the original pentagon with vertices V_1, V_2, V_3, V_4 and V_5 . We need the following simple identity.

$$\frac{1}{2}(\sin \theta_1 + \sin \theta_2) = \sin\left(\frac{\theta_1 + \theta_2}{2}\right) \cos\left(\frac{\theta_1 - \theta_2}{2}\right) \quad (4)$$

The L.H.S. is the area of the figure $OV_1V_2V_3O$. The last factor increases as $|\theta_1 - \theta_2|$ decreases and becomes maximum when $\theta_1 = \theta_2$. So, if $\theta_1 \neq \theta_2$, we let $\theta'_1 = \theta'_2 = \frac{\theta_1 + \theta_2}{2}$. Let the angle bisector of the angle V_1OV_3 meet the unit circle at V'_2 .



Then the area of the figure $OV_1V'_2V_3O$ will be

$$\frac{1}{2}(\sin \theta'_1 + \sin \theta'_2) = \sin\left(\frac{\theta'_1 + \theta'_2}{2}\right) \cos 0 \quad (5)$$

The R.H.S. of (5) is bigger than that of (4). So the figure $OV_1V'_2V_3O$ has a higher area than the figure $OV_1V_2V_3O$. The other parts of the pentagon remain the same. Thus if the adjacent angles θ_1 and θ_2 are unequal, then the pentagon cannot have maximum area. Applying this reasoning to all adjacent pairs, we get that for a pentagon to have the maximum area all θ 's are equal, i.e. the pentagon is regular. ■

A good problem once the idea of marginal improvement strikes in. There is yet another method which, in terms of sophistication,

ranks in between the method of Lagrange's multipliers (which involves calculus of several variables) and the trigonometric solution we have given (which involves no calculus). The method is based on what is called **Jensen's inequality** and needs the concept of the concavity of a function $y = f(x)$. A function $y = f(x)$ is said to be **concave upward** on an interval $[a, b]$, if for all $x_1, x_2 \in [a, b]$, and $\lambda \in [0, 1]$, we have

$$f((1 - \lambda)x_1 + \lambda x_2) \leq (1 - \lambda)f(x_1) + \lambda f(x_2) \quad (6)$$

Geometrically, this means that the chord joining $(x_1, f(x_1))$ to $(x_2, f(x_2))$ lies above the graph of $y = f(x)$ over $[a, b]$. Concave downwards is defined by reversing the inequality. (A function which is concave downward is also called a convex function.) When the inequality is strict for every $\lambda \in (0, 1)$, we add the adverb 'strictly'. Like monotonicity of a function, the definition of concavity makes no reference to anything from calculus. However, just as the sign of the derivative of a function over an interval provides a convenient tool (because of Lagrange's Mean Value Theorem) to prove monotonicity, the sign of the second derivative implies whether a twice differentiable function is concave upward/downward over an interval. Specifically, a function $f(x)$ is concave/strictly concave upward on $[a, b]$ if $f''(x) \geq 0$ for all $x \in (a, b)$ (respectively, $f''(x) > 0$). Similarly $f''(x) \leq 0$ implies convexity and strict convexity if $f''(x) < 0$.

Jensen's inequality extends (6) to any convex combination of points in $[a, b]$. Usually, only a special case of it is needed, which says that if $f(x)$ is strictly concave downward on $[a, b]$, then for every positive integer m , and for every $x_1, x_2, \dots, x_m \in [a, b]$,

$$mf\left(\frac{x_1 + x_2 + \dots + x_m}{m}\right) \geq f(x_1) + f(x_2) + \dots + f(x_m) \quad (7)$$

with equality holding if and only if all x_i 's are equal.

Returning to the problem about the pentagon, let $\theta_1, \dots, \theta_5$ be as above. Then no θ_i can exceed π as otherwise one triangle formed by O and two adjacent vertices will overlap others. Hence $\theta_1, \dots, \theta_5 \in (0, \pi)$ over which the function $f(\theta) = 2 \sin \theta$ is strictly concave since $f''(\theta) = -2 \sin \theta < 0$ for $\theta \in (0, \pi)$. Hence by (2) and Jensen's inequality (with

$m = 5$), we have

$$\frac{5}{2} \sin\left(\frac{2\pi}{5}\right) \geq \frac{1}{2}(\sin \theta_1 + \sin \theta_2 + \dots + \sin \theta_5) \quad (8)$$

with equality holding only when $\theta_i = \frac{2\pi}{5}$ for $i = 1, 2, \dots, 5$, i.e. when the pentagon is regular.

Q.6 Prove that the family of curves

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$$

satisfies

$$\frac{dy}{dx}(a^2 - b^2) = (x + y \frac{dy}{dx})(x \frac{dy}{dx} - y).$$

Solution and Comments: The given family has λ as a parameter. For this reason, such families are called one parameter families of curves. To get the common differential equation satisfied by its members, we differentiate w.r.t. x to get

$$\frac{x}{a^2 + \lambda} + \frac{y \frac{dy}{dx}}{b^2 + \lambda} = 0 \quad (1)$$

The desired differential equation can be obtained by eliminating λ between the equation of the family and (1). For this, we first subtract x times (1) from the given equation of the family to get

$$\frac{y^2}{b^2 + \lambda} - \frac{xy \frac{dy}{dx}}{b^2 + \lambda} = 1 \quad (2)$$

i.e.

$$b^2 + \lambda = y^2 - xy \frac{dy}{dx} \quad (3)$$

Effectively, we have eliminated a . We can similarly eliminate b by first writing

$$\frac{x^2 \frac{dy}{dx} - xy}{a^2 + \lambda} = \frac{dy}{dx} \quad (4)$$

so as to get

$$a^2 + \lambda = \frac{x^2 \frac{dy}{dx} - xy}{\frac{dy}{dx}} \quad (5)$$

Subtracting (3) from (5), we get

$$a^2 - b^2 = \frac{x^2 \frac{dy}{dx} - xy}{\frac{dy}{dx}} - y^2 + xy \frac{dy}{dx} \quad (6)$$

whence

$$\frac{dy}{dx}(a^2 - b^2) = x^2 \frac{dy}{dx} - xy - y^2 \frac{dy}{dx} + xy \left(\frac{dy}{dx}\right)^2 \quad (7)$$

The result follows by factorising the R.H.S. as $(x + y \frac{dy}{dx})(x \frac{dy}{dx} - y)$. ■

A simple but rather laborious problem.

Q.7 Consider a right angled triangle with integer valued sides $a < b < c$ where a, b, c are pairwise coprime. Let $d = c - b$. Suppose d divides a . Then

- (a) Prove that $d \leq 2$.
- (b) Find all such triangles (i.e. all possible triplets a, b, c) with perimeter less than 100.

Solution and Comments: Since the triangle is right angled with hypotenuse c , by the Pythagoras theorem we have

$$a^2 = c^2 - b^2 = (c - b)(c + b) = d(c + b) \quad (1)$$

It is given that d divides a . So we can write $a = kd$ for some positive integer k . Then (1) gives $k^2 d^2 = d(b + c)$ and hence

$$k^2 d = c + b \quad (2)$$

Thus d divides $c + b$. But d also divides $c - b$. (In fact, it equals $c - b$.) So d divides both $c + b + (c - b)$ and $c + b - (c - b)$. That is d divides

both $2c$ and $2b$. As b and c are relatively prime, the only (positive) factor they have in common is 1. Therefore the only common factor of $2b$ and $2c$ is 1 or 2. Hence we must have $d \leq 2$. This proves (a).

For (b), we note first that c is less than half the perimeter. Hence

$$c \leq 49 \quad (3)$$

We have to find all Pythagorean triplets (a, b, c) for which the given conditions hold along with the additional restriction (3). This can be done by trying all possible values of c from 5 to 49. A short cut is provided by the fact that in any Pythagorean triplet (a, b, c) , in which a, b, c have no common factor, the hypotenuse c is of the form

$$c = u^2 + v^2 \quad (4)$$

where u, v are positive integers and $u > v$. The other two sides are $2uv$ and $u^2 - v^2$. The smaller one a can be either $2uv$ (as in the case of the triplet $(8, 15, 17)$ where $u = 4, v = 1$) or $u^2 - v^2$ (as in the case of the triplet $(3, 4, 5)$ where $u = 2, v = 1$). But c always equals $u^2 + v^2$.

This reduces the search for c to those integers from 5 to 49 which can be expressed as a sum of perfect squares $u^2 + v^2$ with $u > v$. Further, as we want a, b, c to be pairwise coprime, we must also have u and v relatively prime. A more sophisticated result which is useful is that 2 and any prime of the form $4k + 1$ can be expressed as a sum of two perfect squares while no other prime can. The product of several primes can be expressed as a sum of perfect squares if and only if in its prime factorisation, every prime of the form $4k + 3$ appears with an even exponent. For example, $45 = 3^2 \cdot 5 = 3^2(1 + 4) = 9 + 36 = 3^2 + 6^2$. Note, however, that 3 and 6 are not relatively prime.

If we apply this sophisticated result, the only possible candidates for c in the range 1 to 49 are: 5, 13, 17, 25, 29, 37 and 41. The corresponding triplets are $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(20, 21, 29)$, $(12, 35, 37)$ and $(9, 40, 41)$. However, $(20, 21, 29)$ has to be discarded as since $d = c - b = 29 - 21 = 8$ does not divide $a = 20$. ■

Without this characterisation of primes of the form $u^2 + v^2$, the search can be shortened using (a) of the problem. If $d = 1$, then the possible values of a are 3, 5, 7, 9 all of which are permissible. If $d = 2$,

the possible values of a are 2,4,6,8,10 and 12. Of these, only 8 and 12 are possible.

The first part of the problem is simple. The second is mostly clerical if attempted exhaustively. The characterisation of primes which can be expressed as sums of perfect squares is due to **Fermat**. It is fairly popular although the proof is non-trivial. Candidates who happen to know it (without proof) stand to gain over those who don't. But considering that 100 is a small number, the advantage is not too unfair.

Q.8 A finite sequence of numbers $(a_1, a_2, \dots, a_n)$ is said to be *alternating* if

$$a_1 > a_2, a_2 < a_3, a_3 > a_4, a_4 < a_5, \dots$$

$$\text{or } a_1 < a_2, a_2 > a_3, a_3 < a_4, a_4 > a_5, \dots$$

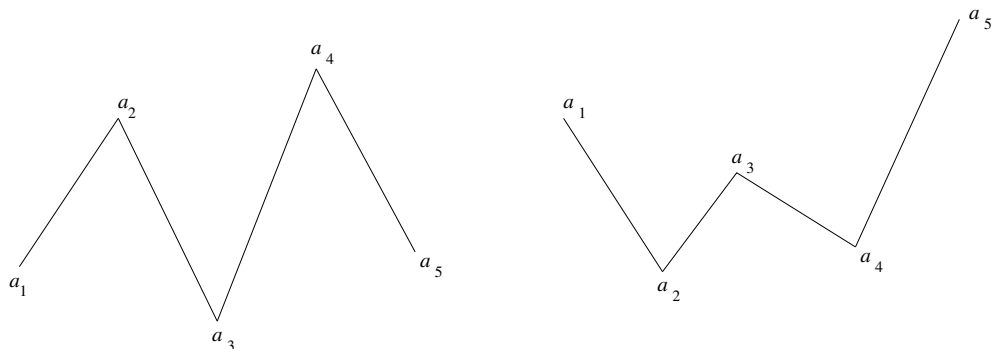
How many alternating sequences of length 5, with *distinct* numbers $a_1, \dots, a_5$ can be formed such that $a_i \in \{1, 2, \dots, 20\}$ for $i = 1, \dots, 5$?

Solution and Comments: For any collection of 5 distinct numbers, the number of alternating sequences of length 5 that can be formed from it is the same, as it does not depend upon which particular numbers appear in the collection. So the first part of the solution is to count the number of selections of 5 distinct number from the set $\{1, 2, \dots, 20\}$.

The answer is $\binom{20}{5}$.

The next part of the problem is to count how many alternating sequences of length 5 can be formed from $\{a_1, a_2, a_3, a_4, a_5\}$ for given 5 distinct real numbers $a_1, \dots, a_5$. Any sequence of length 5 can be looked at as a function defined on the domain set $\{1, 2, 3, 4, 5\}$. The graph of any such function has only 5 points on it, of the form (i, a_i) for $i = 1, \dots, 5$. But if we join (i, a_i) to $(i+1, a_{i+1})$ for $i = 1, 2, 3, 4$ by straight line segments, then we get a vivid idea of the order relationship between two consecutive terms of the sequence. A sequence is alternating if and only if the slopes of the segments are of alternate signs.

Such sequences fall into two kinds, those of the *M*-type and the *W*-type depending upon whether $a_2 > a_1$ or $a_1 > a_2$. These names become suggestive from the following diagrams.



(a) *M*-type sequence

(b) *W*-type sequence

Now suppose, without loss of generality, that $a_1, \dots, a_5$ come from the set $\{1, \dots, 5\}$. In an *M*-type sequence such as the one shown in the figure above, a_2 and a_4 must be the two topmost elements 4 and 5. But they can occur in either order. The elements a_1, a_2, a_3 must be from $\{1, 2, 3\}$ and can appear in any of the $3! = 6$ orders. Thus the number of *M*-type sequences of this kind is $2 \times 6 = 12$. But here we are tacitly assuming that $a_2 > a_5$. It may happen that $a_2 < a_5$. For this to happen, a_4 must be 5 and a_4 1. Now a_2 has to be 3. a_1 and a_3 are 1 and 2 in either order. So there are two additional *M*-type sequences. Similarly there are two additional *M*-type sequences in which $a_1 = 4$. Hence the total count of all alternating *M*-type sequences is thus $12 + 4 = 16$.

By a similar reasoning (or by taking a reflection into the y -axis), the number of *W*-type sequences is also 16. Hence for any fixed subset S of 5 elements from $\{1, 2, \dots, 20\}$ there are 32 alternating sequences that can be formed from S . As S itself can be formed in $\binom{20}{5}$ ways, the total number of alternating sequence of length 5 that can be formed elements from the set $\{1, 2, \dots\}$ is $\binom{20}{5} 32$. ■

A simple problem where the counting is to be done by resolving into various cases, unlike in Q.4 or Q.9 of Paper 1 where it had to be done in one slot. Many candidates are likely to miss the additional four *M*-type sequences. Such problems therefore properly belong to Paper 1, where the multiplicity of options provides some alert.

CONCLUDING REMARKS

Paper 1 problems this year are somewhat better and offer more variety than those last year. For example, Q.5 (finding the number of solutions to the equation $e^x = \sin x$), Q.8 (involving triangular numbers), Q.9 (counting the number of surjective and monotonically increasing functions), Q.10 (asking for the number of roots of a complex polynomial that lie on the unit circle), Q.12 (extracting a subseries of a geometric series), Q.14 (counting the number of subsets of $\{1, 2, \dots, n\}$ that contain no two consecutive numbers), Q.15 (alternately deleting numbers on a circle), Q.20 (alphabetical ordering of permutations), Q.26 (getting a triangle with no interior point from a certain set) and Q.27 (about the validity of a clinical test).

Q.7 (about the implications of a non-negative derivative) and Q.22 (about two functions whose sum and product are differentiable) are very good tests of the ability to give theoretical reasoning. But they are wasted in a multiple choice format, especially when there is no negative marking. Q.26 is also diluted when the construction needed to justify the correct answer does not have to be given. Such questions properly belongs to Paper 2.

The liberal policy of foretelling the candidates that only one option is correct and the lack of care in designing the problems so that the correct option is not easy to guess makes some questions lose steam. For example, Q.1 (testing the points of discontinuity of $|x| + |x + 1| + 3^x$) or Q.24 (about the sum of the largest elements of the subsets of a set) or Q.30 (summing a finite series involving binomial coefficients).

There is also some duplication of ideas. Thus both Q.6 and Q.13 are based on approximating an integral by Riemann sums. Both Q.18 and Q.28 deal with solutions of equations involving the integer part function. Some of this duplication could have been avoided to make room for some topics like inequalities or solutions of triangles which have no representation.

Most of the remaining questions in Paper 1 are routine. In a competitive test with severe time constraint, clubbing them with thought oriented questions gives a distinct advantage to the mediocre candidates over those that are scrupulous.

In Paper 2 the ball game is different because the candidates have to justify their statements. Q.1 (about finding a polynomial having $i + \omega$ as a root), Q.2 (about double roots of a polynomial) and Q.4 (about logarithmic growth being smaller than algebraic growth) are good questions. The locus question (Q.3) is simple but its statement should have made it clear that the candi-

dates must specify exactly which part of the line forms the locus, because this is a departure from an age old practice. Q.6 (finding the differential equation of a one parameter family of ellipses) is absolutely mechanical. Q.5 can be answered quickly using the method of Lagrange's multipliers. Although it is outside the HSC syllabus, many students happen to know it. Unless it is made clear that such advanced methods are not allowed, such students stand to lose. The first part of Q.7 (about Pythagorean triplets) is the only question about number theory. The second part is purely clerical for those who do not know Fermat's two square theorem. Again, to be fair to all, a uniform policy regarding permissibility of its usage is needed to be made known to the candidates well in advance. Q.8 of Paper 2 belongs more properly to Paper 1.

Summing up, both Paper 1 and Paper 2 have a fair percentage of good problems. But the preponderance of mediocre problems in Paper 1 raises some concerns about the efficacy of the good problems. It is ridiculous to give equal weightage to Q.11 (asking for the number of real roots of a factorised polynomial) and to, say, Q.22 about two functions whose sum as well as product are differentiable. Q. 17 (about finding the side of a square) and Q.18 (solving an equation involving the integral part of a number) of Paper 1 are direct replicas of Q.11 and Q.7 respectively from Paper 1 of ISI 2014 Entrance Test. Looks like the pressure of designing as many as 30 questions in Paper 1 is taking the toll on the paper setters! Perhaps it is time that ISI takes a cue from the Chennai Mathematical Institute Entrance Test which has a few well chosen problems and has created a reputation for itself within a short span.